ANSWER KEY

English Test

1. A	21. D	41. B	61. C
2. H	22. F	42. H	62. G
3. B	23. D	43. D	63. C
4. H	24. G	44. H	64. F
5. D	25. B	45. A	65. B
6. F	26. H	46. G	66. J
7. B	27. C	47. A	67. B
8. J	28. F	48. H	68. J
9. B	29. A	49. C	69. C
10. J	30. G	50. J	70. G
11. C	31. D	51. D	71. D
12. F	32. H	52. J	72. F
13. C	33. B	53. B	73. C
14. G	34. G	54. F	74. F
15. D	35. A	55. C	75. D
16. G	36. H	56. F	
17. D	37. D	57. D	
18. F	38. J	58. G	
19. C	39. C	59. A	
20. H	40. F	60. H	

Mathematics Test

1. B	21. C	41. D
2. J	22. K	42. F
3. C	23. B	43. B
4. J	24. H	44. K
5. C	25. D	45. B
6. F	26. G	46. H
7. E	27. D	47. E
8. H	28. G	48. K
9. D	29. A	49. C
10. G	30. J	50. G
11. C	31. E	51. C
12. F	32. G	52. G
13. D	33. D	53. B
14. J	34. G	54. K
15. B	35. C	55. D
16. F	36. H	56. F
17. C	37. B	57. E
18. J	38. J	58. F
19. A	39. C	59. A
20. G	40. J	60. K

Reading Test

1. A	21. A
2. J	22. G
3. C	23. B
4. J	24. H
5. A	25. C
6. H	26. J
7. B	27. C
8. J	28. F
9. D	29. C
10. G	30. F
11. B	31. B
12. H	32. F
13. A	33. D
14. J	34. J
15. C	35. C
16. H	36. F
17. A	37. B
18. G	38. F
19. A	39. C
20. H	40. G

Science Reasoning Test

1. C	21. B
2. H	22. J
3. A	23. A
4. G	24. F
5. A	25. D
6. F	26. G
7. C	27. D
8. J	28. G
9. B	29. D
10. G	30. F
11. A	31. C
12. G	32. F
13. A	33. A
14. H	34. H
15. D	35. D
16. H	36. H
17. D	37. A
18. G	38. G
19. C	39. B
20. H	40. J

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SCORING GUIDE

Your final reported score is your COMPOSITE SCORE. Your COMPOSITE SCORE is the average of all of your SCALE SCORES.

Your SCALE SCORES for the four multiple-choice sections are derived from the Scoring Table on the next page. Use your RAW SCORE, or the number of questions that you answered correctly for each section, to determine your SCALE SCORE. If you got a RAW SCORE of 60 on the English test, for example, you correctly answered 60 out of 75 questions.

Step 1 Determine your RAW SCORE for each of the four multiple-choice sections:

English _____

Mathematics _____

Reading _____

Science Reasoning _____

The following Raw Score Table shows the total possible points for each section.

RAW SCORE TABLE	
KNOWLEDGE AND SKILL AREAS	**RAW SCORES**
ENGLISH	75
MATHEMATICS	60
READING	40
SCIENCE REASONING	40
WRITING	12

Multiple-Choice Scoring Worksheet

Step 2 Determine your SCALE SCORE for each of the four multiple-choice sections using the following Scoring Worksheet. Each SCALE SCORE should be rounded to the nearest number according to normal rules. For example, 31.2 ≈ 31 and 31.5 ≈ 32. If you answered 61 questions correctly on the English section, for example, your SCALE SCORE would be 29.

English	_____	× **36** =	_____	÷ **75** =	_____		
	RAW SCORE					**− 2**	(*correction factor)
					_____		
					SCALE SCORE		
Mathematics	_____	× **36** =	_____	÷ **60** =	_____		
	RAW SCORE					**+ 1**	(*correction factor)
					_____		
					SCALE SCORE		
Reading	_____	× **36** =	_____	÷ **40** =	_____		
	RAW SCORE					**+ 2**	(*correction factor)
					_____		
					SCALE SCORE		
Science Reasoning	_____	× **36** =	_____	÷ **40** =	_____		
	RAW SCORE					**+ 1.5**	(*correction factor)
					_____		
					SCALE SCORE		

*The correction factor is an approximation based on the average from several recent ACT tests. It is most valid for scores in the middle 50% (approximately 16–24 scale composite score) of the scoring range.

The scores are all approximate. Actual ACT scoring scales vary from one administration to the next based upon several factors.

If you take the optional Writing Test, you will need to combine your English and Writing scores to obtain your final COMPOSITE SCORE. Once you have determined a score for your essay out of 12 possible points, you will need to determine your ENGLISH/WRITING SCALE SCORE, using both your ENGLISH SCALE SCORE and your WRITING TEST SCORE. The combination of the two scores will give you an ENGLISH/WRITING SCALE SCORE, from 1 to 36, that will be used to determine your COMPOSITE SCORE mentioned earlier.

Using the English/Writing Scoring Table, find your ENGLISH SCALE SCORE on the left or right hand side of the table and your WRITING TEST SCORE on the top of the table. Follow your ENGLISH SCALE SCORE over and your WRITING TEST SCORE down until the two columns meet at a number. This number is your ENGLISH/WRITING SCALE SCORE and will be used to determine your COMPOSITE SCORE.

Step 3 Determine your ENGLISH/WRITING SCALE SCORE using the English/Writing Scoring Table on the following page:

English _____

Writing _____

English/Writing _____

ENGLISH/WRITING SCORING TABLE

ENGLISH SCALE SCORE	WRITING TEST SCORE											ENGLISH SCALE SCORE
	2	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	
36	26	27	28	29	30	31	32	33	34	32	36	**36**
35	26	27	28	29	30	31	31	32	33	34	35	**35**
34	25	26	27	28	29	30	31	32	33	34	35	**34**
33	24	25	26	27	28	29	30	31	32	33	34	**33**
32	24	25	25	26	27	28	29	30	31	32	33	**32**
31	23	24	25	26	27	28	29	30	30	31	32	**31**
30	22	23	24	25	26	27	28	29	30	31	32	**30**
29	21	22	23	24	25	26	27	28	29	30	31	**29**
28	21	22	23	24	24	25	26	27	28	29	30	**28**
27	20	21	22	23	24	25	26	27	28	28	29	**27**
26	19	20	21	22	23	24	25	26	27	28	29	**26**
25	18	19	20	21	22	23	24	25	26	27	28	**25**
24	18	19	20	21	22	23	23	24	25	26	27	**24**
23	17	18	19	20	21	22	23	24	25	26	27	**23**
22	16	17	18	19	20	21	22	23	24	25	26	**22**
21	16	17	17	18	19	20	21	22	23	24	25	**21**
20	15	16	17	18	19	20	21	21	22	23	24	**20**
19	14	15	16	17	18	19	20	21	22	23	24	**19**
18	13	14	15	16	17	18	19	20	21	22	23	**18**
17	13	14	15	16	16	17	18	19	20	21	22	**17**
16	12	13	14	15	16	17	18	19	20	20	21	**16**
15	11	12	13	14	15	16	17	18	19	20	21	**15**
14	10	11	12	13	14	15	16	17	18	19	20	**14**
13	10	11	12	13	14	14	15	16	17	18	19	**13**
12	9	10	11	12	13	14	15	16	17	18	19	**12**
11	8	9	10	11	12	13	14	15	16	17	18	**11**
10	8	9	9	10	11	12	13	14	15	16	17	**10**
9	7	8	9	10	11	12	13	13	14	15	16	**9**
8	6	7	8	9	10	11	12	13	14	15	16	**8**
7	5	6	7	8	9	10	11	12	13	14	15	**7**
6	5	6	7	7	8	9	10	11	12	13	14	**6**
5	4	5	6	7	8	9	10	11	12	12	13	**5**
4	3	4	5	6	7	8	9	10	11	12	13	**4**
3	2	3	4	5	6	7	8	9	10	11	12	**3**
2	2	3	4	5	6	6	7	8	9	10	11	**2**
1	1	2	3	4	5	6	7	8	9	10	11	**1**

Step 4 Determine your COMPOSITE SCORE by finding the sum of all your SCALE SCORES for each of the four sections: English only (if you do not choose to take the optional Writing Test) *or* English/Writing (if you choose to take the optional Writing Test), Math, Reading, and Science Reasoning, and divide by 4 to find the average. Round your COMPOSITE SCORE according to normal rules. For example, 31.2 ≈ 31 and 31.5 ≈ 32.

_____	+	_____	+	_____	+	_____	=	_____
ENGLISH *OR* ENGLISH/WRITING SCALE SCORE		MATHEMATICS SCALE SCORE		READING SCALE SCORE		SCIENCE SCALE SCORE		SCALE SCORE TOTAL

_____	÷ 4 =	_____
SCALE SCORE TOTAL		COMPOSITE SCORE

ANSWERS AND EXPLANATIONS

English Test Explanations

PASSAGE I

1. **The best answer is A.** The comma is needed to separate the introductory clause from the main clause. The date does not have to be separated by two commas, and a semicolon would indicate two complete sentences.

2. **The best answer is H.** "Far more rarer" is redundant. Remember that the ACT rewards conciseness. In addition, to form the comparative, either add the word "more" or add the "-er," not both. Always choose the grammatically correct response with the fewest words—in this case, answer choice H. Answer choice G is awkward and wordy.

3. **The best answer is B.** The gerund "leading" does not bear tense and so cannot be the main verb of the sentence. In this case, the sentence needs a form of the future tense because the search would occur after the student's report. Answer choice C, "was leading," is in the past. Answer choice D, "was lead," could never be correct because the past form of "lead" is "led" and the passive voice does not make sense here.

4. **The best answer is H.** The sentence would be improved with a definite pronoun to identify the habitat. Answer choices F and J are indefinite, and therefore should be eliminated. "Its" is correct because "woodpecker" is singular. Answer choice G is correspondingly incorrect because "their" is plural.

5. **The best answer is D.** The subject of the passage and of the paragraph is the ivory-billed woodpecker. It's the only "species" designated as such. Therefore, it's redundant to combine "species" with "ivory-billed woodpecker" in this sentence. The ACT values conciseness. Always choose the least-wordy option that is grammatically correct.

6. **The best answer is F.** In this sentence, it is necessary to use commas to set off the introductory phrase and the two independent clauses. Answer choices H and J use commas to separate short phrases that are not full clauses. Semicolons are used between independent clauses not linked by a conjunction, so answer choice G would be correct if there was no conjunction, "and." Therefore, the sentence is best as it stands.

7. **The best answer is B.** Because "species" ends in an "s," an apostrophe can simply be added to the end of the word without adding another "s" to create possession. Answer choice C is too wordy. Answer choice D is wrong for two reasons: first, "species" has to modify "survival," not the other way around, and second, in this example "species" should not be possessive.

8. **The best answer is J.** Answer choice H is incorrect because the passage already states that the most recent sighting was in 1999. The rest of the sentence is too vague to answer the question. Answer choice G does not make clear if the month in 2002 was when the biologists searched the woods or when the bird was spotted again. While answer choice F is grammatically correct and offers new information, choice J adds the most specific details to the passage.

9. **The best answer is B.** A comma is used before a coordinating conjunction joining independent clauses in a sentence. Therefore, answer choice B is correct, and answer choice C is incorrect. Answer choice A adds an extra "it" to the sentence and uses a semicolon where a comma is needed. The clause "later determined," in answer choice D, does not make sense.

10. **The best answer is J.** The topic of the paragraph is the new search for the ivory-billed woodpecker. The search is instigated by the 1999 sighting, so Sentence 1 should remain first. Sentence 2 introduces the search and should continue to follow Sentence 1. The conjunction "though" in Sentence 3, however, indicates a contrast that only makes sense if it comes after the lack of evidence cited in Sentence 4. The fifth sentence starts with the phrase, "In the end," and should remain last.

11. **The best answer is C.** Answer choices A and D are too wordy. Answer choice B is more concise, but would need a semicolon to separate the clauses because they form two complete sentences. Answer choice C uses a period to separate the sentences and is the best answer choice.

12. **The best answer is F.** Commas should be used to separate the elements in a series, but it is not necessary to put a comma between the last element in the series and the following verb. So, answer choices H and J are incorrect. A comma should come between the second to last element and the conjunction, so answer choice G is incorrect.

13. **The best answer is C.** The idea here is that researchers feared a sudden influx of birdwatchers drawn by the new sightings. The only selection that matches that meaning is answer choice C.

14. **The best answer is G.** The word "accepts" makes the most sense in the sentence because it clearly and simply indicates present tense. The phrase "is accepting of" may appear to be correct, but it is rather awkward and would be more appropriate if it referred to evidence of the birds rather than their actual existence.

15. **The best answer is D.** Sentence 3 discusses "sightings" by researchers in Arkansas. This would lead easily to information about those sightings, like the "15 sightings" mentioned in the extra sentence. Sentences 1 and 2 could refer to multiple video tapings, but the tapings are from motion-sensing cameras and are different from personal sightings. Therefore, the best place for the new sentence is after Sentence 3, answer choice D.

PASSAGE II

16. **The best answer is G.** This sentence is made up of two complete clauses that could both stand independently. Therefore, answer choices H and J would be acceptable ways to write the sentence. Answer choice H makes two complete sentences separated by a period, while J uses a semicolon to separate them. Answer choice F, "no change," is also acceptable because the conjunction "and" follows the comma. Answer choice G, however, is not acceptable because it uses a comma without a conjunction to join two complete thoughts. This is called a "comma splice," and it is grammatically incorrect.

17. **The best answer is D.** The sentence is written in the present tense, and therefore, the correct form of the verb is "spend." Answer choice B is also wrong because the sentence has a compound subject ("editors and I") which requires a plural verb form ("are" instead of "am"). Answer choice C is incorrect because it leaves the clause without a main verb.

18. **The best answer is F.** The passage is acceptable as it stands because it doesn't need a causal connection between the end of the first paragraph and the beginning of the second. In other words, the editorial board doesn't reconvene in the summer as a result of the editors' long hours, so it's unnecessary to say "consequently" or "as a result," as in answer choices G and J. "However" (answer choice H) is also unnecessary because there's no contrast.

19. **The best answer is C.** The sentence needs a relative clause to explain what is done with the call for papers. In this case, the clause needs a subject and a verb in the present tense. Answer choice D lacks a subject. Answer choice B is not in standard English word order (i.e. subject, verb, object). And finally, the phrase "nationwide research institutions" in answer choices A and C implies that each institution exists nationwide (as opposed to, say, a state university). Only answer choice C is complete and grammatically correct.

20. **The best answer is H.** "Researchers and authors" forms a compound subject. A compound subject requires a plural verb, which this sentence has. It does not need any commas, however, so the correct answer is answer choice H.

21. **The best answer is D.** This sentence needs a main clause that contains a subject, verb, and object. Answer choices B and C would leave an incomplete sentence. Answer choice A has an unclear antecedent—that is, the sentence does not make clear who is meant by "they." Only answer choice D is clear and grammatically correct.

22. **The best answer is F.** This sentence has two parallel subordinate clauses. Each clause should have the same structure, so the editors should "read" and "submit" each work. Answer choices G and J change the form of the second verb. Answer choice H is incorrect because, as it stands, the second clause is not a complete sentence and cannot be set apart by a semicolon.

23. **The best answer is D.** The sentence as it stands (answer choice F) is grammatically correct. The next sentence begins with the pronoun "those" that refers back to the subject of the previous sentence. In answer choices A and C, the subject is the editors. In answer choice D, the subject is the paper submissions, which is a better match for what comes next. Answer choice B is too wordy and the word "either" is unnecessary.

24. **The best answer is G.** The added sentence would be helpful because it would clarify how a paper might be indecipherable. This is why answer choice G is correct and answer choice J is incorrect. The form of the papers is relevant (answer choice H) because papers can be returned to the author if the formatting is poor. Answer choice F is incorrect because the new sentence does not change or establish the tone—the first sentence has

already done that by introducing the possibility of bad papers.

25. **The best answer is B.** The clause needs a finite verb, not a participle. This eliminates answer choices A and C. Answer choice D is incorrect because a verb should be modified by an adverb ("immediately"), not an adjective ("immediate").

26. **The best answer is H.** Answer choices F and G are essentially interchangeable. They both imply that the most beneficial information for the journal's readers is an example of the professional standards of the journal. Therefore, it is best to eliminate both choices. Answer choice J implies a contrast between the beneficial information and the professional standards. This does not make sense within the passage. Answer choice H, however, suggests that the professional standards refer, at least in part, to the format and presentation mentioned earlier, while the beneficial information is an additional requirement for publication. This makes the best sense in context.

27. **The best answer is C.** The simplest option is often the best. In this case, answer choice C has the fewest clauses and the simplest structure. It is grammatically correct and not redundant.

28. **The best answer is F.** Adverbial clauses, like the one that starts this sentence, need to be separated from the main clause by a comma. This eliminates answer choices H and J. The phrase, "once we finalize," indicates a single action that should be followed by another single action to keep parallel structure. The phrase, "we're sending," implies repeated or continuing action. Therefore, answer choice G is incorrect. Answer choice F has a comma after "layout," and "send" is parallel in construction with "finalize," so this is the best answer choice.

29. **The best answer is A.** The verb "meets" indicates that the action is taking place in the present, while the journals are being prepared to mail. Answer choice B is incorrect because the noun phrase "editorial board" is singular, and so requires a singular verb. Both answer choices C and D create incomplete sentences.

30. **The best answer is G.** In standard English, adjectives typically go before the nouns they modify. In this case, "well-earned" modifies "vacation," and so answer choice G is correct. The other choices are incorrect or simply awkward.

PASSAGE III

31. **The best answer is D.** "Italy," as it is used here, needs to be put in the possessive. Because the country is singular and does not end in an "s," the correct form is "Italy's."

32. **The best answer is H.** The passage states that Montessori was Italy's first modern woman physician. That implies she was trying to advance *within* the profession (answer choice H). Answer choice F would imply that the male physicians were helping her, which is contrary to the statement that she struggled. Answer choice G would imply that she was trying to promote the profession and not her own career, which is not supported by the rest of the passage. In answer choice J, the pronoun does not appear to refer to anything.

33. **The best answer is B.** A colon is used to introduce an explanatory phrase if the first part of the sentence creates a sense of anticipation about what follows. This sentence does not do that, so answer choice A is incorrect. However, the addition does provide important information about why Montessori was sent there, so it would be wrong to eliminate it as in answer choice D. The additional information is in a subordinate clause and so needs a comma to separate it from the rest of the sentence. Answer choice B is correct.

34. **The best answer is G.** The passage makes a distinction between Montessori's training and the processes of the mind. Only answer choice G, "anatomy and physiology," makes this distinction clear. Answer choices F, H, and especially J are broad and could include study of the mind. They do not make a clear distinction and are incorrect.

35. **The best answer is A.** This clause is specifying Montessori's studies in the processes of the mind. It needs to be an example of a process. Answer choices C and D put the emphasis on the children, not on the way they learn. Answer choice B has the right emphasis, but it is unnecessarily wordy.

36. **The best answer is H.** The main clause in this sentence begins with "Montessori established a 'children's house.'" What comes before is an adverbial clause that needs to be separated from the main clause by a comma. This eliminates answer choices G and J, which were also wordy and awkward. Answer choice F is also wordy and difficult to follow. Answer choice H makes clear that Montessori had sixty children from the working class when she established her house.

37. **The best answer is D.** The word "ready" actually refers to *how* the children learn, and so it modifies the verb "learn" not the noun "children." Therefore, it should be in the form of an adverb, "readily." That eliminates answer choices A and C. There is no natural pause between "children" and "readily" and no grammatical need to put a comma, so answer choice B is incorrect.

38. **The best answer is J.** Sentence 2 establishes the topic of the paragraph: Montessori's experiences in her "children's house." Sentences 3 and 4 give examples of what she learned. Sentence 1 describes how these specific experiences influenced her subsequent work. Therefore, sentence 1 should conclude the paragraph and be put after Sentence 4.

39. **The best answer is C.** By mentioning both development and learning, answer choice C combines the information in answer choices A and B. Therefore, it is more complete and specific than either of them. Answer choice D introduces the idea of school districts, which does not belong in this passage.

40. **The best answer is F.** The topic of the sentence is development, not children. This means the child's development is supposed to be compared to standards and norms, not the child in general. This eliminates answer choices H and J. Answer choice G places the apostrophe in the wrong place. The correct placement is used in answer choice F.

41. **The best answer is B.** This sentence has a compound verb: the subject "children" is shared by "develop" and "acquire." Commas are not used to separate compound verbs. (Consider the related rule about compound subjects: *Bill, and I went to the store* is clearly incorrect.) This eliminates answer choices A and C. Meanwhile, semicolons are used to separate two complete sentences. The final clause is not a complete sentence (it lacks a subject), so answer choice D is incorrect.

42. **The best answer is H.** To maintain parallel construction with the verb form "provides," use the present tense "observes."

43. **The best answer is D.** The second clause in this sentence is a relative clause. That means it is a subordinate clause that behaves like an adjective in that it helps describe a noun phrase. In this case, the noun phrase is the fundamental processes of the Montessori Method. The subsequent clause provides more information about that noun phrase and so acts like an adjective. Relative clauses are introduced by relative pronouns. The only choice of a relative pronoun is "which," answer choice D.

44. **The best answer is H.** In simple sentences, standard English usually puts the adjective in front of its noun. While answer choice F is not grammatically incorrect, it does not match the straightforward style of the rest of the passage. Answer choice G has unnecessary commas, while J is awkward and too wordy.

45. **The best answer is A.** The fourth paragraph shifts the passage from a historical discussion of how Montessori developed her method to a description of how Montessori's method is used by schools today. This eliminates answer choice B, which talks about past development, and answer choice D, which ignores the shift in topic. Answer choice C is wrong because a discussion of Montessori's current significance shows her importance to her field, which helps justify a biography of her in the first place. That leaves answer choice A, which is correct.

PASSAGE IV

46. **The best answer is G.** The verb "immerse" needs to agree with the preceding verb "want" in order to maintain parallel construction. The verb "want" is actually "will . . . want" and is in the future tense. Therefore, answer choice G, which creates "you will immerse," is the correct answer.

47. **The best answer is A.** The underlined portion forms a clause that contrasts ("but") Paris' awe-inspiring cultural treasures with the ease of travel on the Metro. Answer choice A works grammatically because it creates two independent clauses separated by a comma. To be correct, answer choice B would need a period or a semicolon after "treasures" because "France's capital has cultural treasures" is a complete sentence. Both answer choices C and D are noun phrases in need of a verb to make sense.

48. **The best answer is H.** The underlined section introduces a nonrestrictive relative clause, which means it contributes information about what is referenced (the subway) without restricting the reference. In this case, the clause tells the reader the name of the subway system but does not distinguish it, say from subways in other cities. Nonrestrictive relative clauses are introduced by "wh"-relative pronouns (who, which, etc.) and are frequently

separated from the rest of the sentence by a comma. Answer choice G is awkward and ungrammatical. Answer choice F uses the wrong pronoun, while answer choice J would only be correct if the final clause were a complete sentence.

49. The best answer is C. The gerund "making" marks the beginning of a descriptive phrase that needs to be set apart by some form of punctuation. A dash indicates a sharp break in the continuity of the sentence. A semicolon indicates the start of a new complete sentence. Neither of these occur in this sentence, so a comma is the best choice.

50. The best answer is J. The sentence needs a main verb, and "has" fits the sense well. While it is possible to contract the subject/verb "Metro has" to "Metro's," the apostrophe is in the wrong place in answer choice G and missing in answer choice F. Answer choice H is incorrect because the Metro is more than just stations; it is a system of travel.

51. The best answer is D. Answer choice D illustrates how easy it is to access the Metro from anywhere in Paris. This would provide a good example of the Metro system's expansiveness and coverage. Answer choice B would contradict the desired emphasis. While answer choice A would indicate the Metro is popular, it doesn't provide proof that the stations are numerous. Answer choice C is irrelevant.

52. The best answer is J. The phrase "ideal for tourist travel" is all-encompassing. It could be modified by a phrase that restricts the ideal (for example, "unless you don't speak French"), but none of the choices does that. Therefore, it is best to omit the underlined portion altogether.

53. The best answer is B. Answer choice B correctly states that the stations took their names from the adjacent attractions. Answer choice A implies that the attractions named the stations, which would be impossible because the attractions are inanimate. Answer choice C implies the opposite, which is also impossible. Answer choice D makes no sense.

54. The best answer is F. Here, we have a compound adjective. The author loves the feeling of going underground and coming back to the surface. The going and coming should not be separated by a comma. "Coming" should also remain the gerund form to maintain parallel construction with "going."

55. The best answer is C. The verb "to swipe" should be in the imperative form because the author is giving a directive. The imperative form of "to swipe" is "swipe."

56. The best answer is F. Answer choice F emphasizes how a rider can utilize the full extent of the Metro at a very reasonable price. This definitely illustrates a tremendous advantage to using the subway system. Answer choice H does this, too, but to a lesser extent. Answer choice G is even more vague, and therefore, less acceptable as an answer choice. Answer choice J points out a disadvantage to using the Metro and should be eliminated.

57. The best answer is D. Commas are used to separate words or phrases in a series of three or more items. ACT style always uses a comma between the second-to-last item and the conjunction (usually "and"). Answer choices B and C do not have enough commas, while answer choice A adds an unnecessary "and" before "parking."

58. The best answer is G. In standard written English, the modifier usually goes before whatever it is modifying. Here, "generally" modifies the adjective "accustomed" and so should go before it. The main verb "to be" should also agree in tense with the other main verbs. The other verbs are in the present tense, so this sentence should use the form "are." Both of these points support answer choice G. Because "accustomed" is an adjective, there is no need to change its spelling, answer choice J.

59. The best answer is A. The author tells the reader that the Metro provides incredible freedom of movement. Because it's clear that the author is talking about the Metro, it is redundant to use the phrase "Paris Metro" again, as in answer choices B and C. Answer choice B could also only be correct if the relative pronoun "that" were removed. Answer choice D would leave a clause that reads "that providing it." This doesn't make sense because it needs something to come next. Therefore, answer choice D should be eliminated.

60. The best answer is H. Paragraph 3 talks about the ease of traveling around the city and the joys of finding new places. The new sentence continues the idea of easy traveling but adds the delight of returning to a familiar place. This completes the circuit of a day and leads smoothly to the last sentence: "Metro travel couldn't be easier!"

PASSAGE V

61. The best answer is C. *Plentiful* and *abundant* are synonyms. Therefore, it is redundant to use both of them in this sentence. Answer choice D is also grammatically incorrect.

62. The best answer is G. This sentence contains only one clause. Therefore, it is unnecessary to divide it with punctuation of any kind. The remaining answer choices imply strong divisions that are not supported by the structure of the sentence.

63. The best answer is C. Here it is necessary to choose the correct preposition to introduce the rest of the prepositional phrase, "the acidic environment of these marshlands." The preposition "in" suggests location—that the stuff that makes up peat is kept from decomposing because it is in the acidic environment. "With" would suggest accompaniment—that the stuff is sitting in the bog alongside the acidic environment, which is not correct. "For" is incorrect because it suggests that some other agent is preventing decomposition to maintain an acidic environment. Finally, "among" would only be used if there was a plural object in the prepositional phrase.

64. The best answer is F. "When peat is harvested" is an introductory clause. It should be separated from the main clause by a comma. A period or semicolon, as in answer choices G and H, would mean that it is a complete sentence.

65. The best answer is B. Choosing the proper verb form depends on understanding the sense of the sentence. Something is happening to the peat in the Irish villages. Peat is dug out of wet marshlands and used as fuel. It makes sense that at some point freshly dug peat has to dry out. This process would have to occur over time, so the correct verb form would use the participle "drying" to indicate action over time. This is called the progressive aspect.

66. The best answer is J. The passage's main idea is the use of peat as a fuel source. The most appropriate and relevant contrast would be in direct opposition to that main idea. Answer choice F talks about the amount of peat used, while Answer choices G and H state where it is used. These are details that add to the main idea, not contrast with it. Only answer choice J offers alternatives to peat as a fuel source. Therefore, answer choice J is the correct answer.

67. The best answer is B. First, the verb in this clause refers to the peat, which is singular. Therefore, the verb form must agree with a singular subject. That eliminates answer choice A. Also, the verb needs to be in the past tense to be consistent with "plowed" in the preceding clause.

68. The best answer is J. The underlined section introduces a nonrestrictive relative clause, which means it contributes information about what is referenced (Bord Na Móna) without restricting the reference. In this case, the clause tells the reader the output of the company but does not distinguish it, say, from the output of other companies. Nonrestrictive relative clauses are introduced by "wh"-relative pronouns (who, which, etc.) and are separated from the rest of the sentence by a comma. The relative clause is not a complete sentence, so answer choices F and G could never be correct.

69. The best answer is C. Answer choice C has what is known as a "comma splice." This means that the sentence is composed of two complete sentences joined together by a comma with no separating preposition or conjunction. Answer choice A has the adverb conjunction "while" after the comma, and answer choice D has the word "and." Both choices are grammatically correct and acceptable. Answer choice B simply separates the two sentences, and so it, too, is acceptable.

70. The best answer is G. Answer choice G is the most concise option. It is also consistent with "ridging machine" used later in the paragraph. In fact, none of the equipment is defined or described in any detail. Therefore, "harrow machine" fits best with the language of the paragraph.

71. The best answer is D. The correct form of the verb is "passes" to stay consistent with the rest of the paragraph. The word "passes" is used several times in the paragraph, and this sentence gives no reason to change tenses.

72. The best answer is F. Answer choices G and H would create a comma splice by combining two complete sentences with only a comma. Answer choice F uses the passive form of the verb "to draw" to correctly describe the harvester as putting the peat into a collection bin. Answer choice J is incorrect because it uses the active form of the verb. This implies that the peat is putting something into the bin, which does not make any sense in the context of the paragraph.

73. The best answer is C. The phrase "the history" is too vague. The sentence would have better clues for the reader with a pronoun referring to the

sentence's subject, "peat." The correct pronoun is "its." Answer choice B, "it's," is a contraction of "it is." The verb is not necessary here, and answer choice B is incorrect.

74. The best answer is F. The passage puts peat in the context of Ireland from the very beginning. Peat is not just a resource, it is an *Irish* resource. Answer choice F is the only choice that acknowledges peat's role in Irish culture. Therefore, it provides the best link between the opening and closing sentences.

75. The best answer is D. An essay on the evolution of power sources suggests that the topic would have some breadth. In other words, it would cover more than one type of power source and more than one country. This passage only describes one power source and locates that source in only one country. Therefore, answer choice D is correct.

Mathematics Test Explanations

1. **The correct answer is B.** To find how many degrees Fahrenheit the temperature increased, you need to find the range between the highest and lowest temperature for the day. To do so, subtract the lowest temperature from the highest temperature: $21 - (-7) = 21 + 7 = 28$ degrees Fahrenheit. If you chose D, you may have forgotten that subtracting a negative number is the same thing as adding that number.

2. **The correct answer is J.** To find how much money you save when you buy a \$12 video that is on sale for 20% off, you should find 20% of \$12. To do so, multiply \$12 by 20%, or 0.20; (\$12)(0.20) = \$2.40.

3. **The correct answer is C.** To compute the average of the four prices per gallon of gasoline, add the four prices and divide that sum by the number of prices, which in this case is four: $\frac{\$2.45 + \$2.50 + \$2.49 + \$2.56}{4} = \frac{10}{4}$, \$2.50.

4. **The correct answer is J.** The volume of a cube is calculated by multiplying the length by the width by the height. Because each side of a cube is congruent, simply calculate $(\text{side})^3$. Therefore, the volume is 5^3, or 125.

5. **The correct answer is C.** Given that $3(a-6) = -21$:

$$\frac{3(a-6)}{3} = \frac{-21}{3}$$

$$(a-6) + 6 = -7 + 6$$

$$a = -1$$

6. **The correct answer is F.** When a cantaloupe weighing 3.0 pound cantaloupe costs \$3.87, you can find the cost of a 2.25-pound cantaloupe by setting up a proportion, or by finding the cost per pound and then multiplying by the weight. To solve with a proportion, let x be the cost of the 2.25 pound cantaloupe. Thus $\frac{x}{2.25} = \frac{\$3.87}{3.0}$; solving for x yields $x = \frac{\$3.87 \times 2.25}{3.0} = \$2.90$.

   To solve by finding the per pound rate, divide \$3.87 by 3.0 to find the cost per pound in the first cantaloupe, and then multiply that rate by 2.25 pounds, the weight of the second cantaloupe:

$$\frac{\$3.87}{3.0} \text{ pounds} = \$1.29 \text{ per pound}$$

$$(\$1.29 \text{ per pound})(2.25 \text{ pounds})$$
$$= \$2.90$$

7. **The correct answer is E.** Remembering that perpendicular lines form right angles, the angles *CDA* and *CDB* both measure 90° and thus are equal in measure, answer choice E. From the figure it may appear as if segments *AD* and *DB* are equal in length, but that cannot be determined based solely on the information given in the problem.

8. **The correct answer is H.** To solve this problem, since the equations equal each other, you must combine like terms and solve for x:

$$6x - 5 = 3x - 16$$

$$6x - 5 - 3x = 3x - 16 - 3x$$

$$3x - 5 + 5 = -16 + 5$$

$$\frac{3x}{3} = -\frac{11}{3}$$

$$x = -\frac{11}{3}$$

9. **The correct answer is D.** To solve this problem, you must distribute correctly and remember the proper signs. When you distribute $y(3-y)+5(y-7)$, you arrive at $3y - y^2 + 5y - 35$. Once you add like terms, you get $-y^2 + 8y - 35$.

10. **The correct answer is G.** First, find the value of s. Since there are 8 arcs of each length, it follows that there are 8 arcs of length $(4 + s)$. Since the circumference is 40, then $8(4 + s) = 40$. Solve for s: $32 + 8s = 40$ and $s = 1$. Now that you know the value of each length, you can solve for the degree measure of each of the arcs of length s. You know that the entire circle has 360°. You also know that the length of the smaller arc (s) is $\frac{1}{4}$ the length of the larger arc. Therefore, you can set up the following equation: $8x + 8\frac{1}{4}x = 360°$, where x is the degree measure corresponding to each arc of length 4 and $\frac{1}{4}x$ is the degree measure corresponding to each arc of length $s = 1$. Now you have $8x + 8\frac{1}{4}x$, or $8x + 2x$, which is $10x = 360°$. So $x = 36°$. Recall that x is the degree measure corresponding to each arc of length 4. The degree measure corresponding to each arc of length s is $\frac{1}{4}x$; $\frac{1}{4}(36) = 9°$.

11. The correct answer is C. To solve this problem, divide the number of those people in favor of constructing a new high school by the total number of people polled and simplify, as follows:

$$44 \div (44 + 58 + 8) = 44/110 = 2/5$$

12. The correct answer is F. To determine the ratios, it is helpful to assign numerical values to the lengths of the segments. In this case, since the ratio of AB to BC is 1:4, let AB have a length of 1 and BC have a length of 4. Since $AC = AB + BC$, it can be given a length of 5. Therefore, the ratio of lengths AB to AC is 1:5.

13. The correct answer is D. To find half of 9 feet 6 inches, recall that there are 12 inches in a foot. Therefore, 9 feet 6 inches is equal to $9(12) + 6 = 114$ inches. Half of 114 inches is 57 inches, which is equivalent to 4 feet 9 inches $[4(12) + 9 = 57]$.

14. The correct answer is J. If t is the speed of the truck, then twice the speed of the truck can be written as $2t$. If the speed of the car exceeded twice the speed of the truck by 15 mph, the car's speed was $2t + 15$.

15. The correct answer is B. If the length of the chord XY is 8, then the length of the segment XZ is 4. The right triangle XCZ has a hypotenuse of 5 (equal to the radius), leg of 4, and second leg equal to the length of segment CZ. To find the length of segment CZ you can use the Pythagorean Theorem for a right triangle ($c^2 = a^2 + b^2$ where c is the hypotenuse and a and b are legs):

$$5^2 = 4^2 + CZ^2$$
$$CZ^2 = 5^2 - 4^2$$
$$CZ^2 = 25 - 16$$
$$CZ^2 = 9$$
$$CZ = 3$$

16. The correct answer is F. To find the value of the expression $2x^3 - x^2 + 3x + 5$ for $x = -2$, simply substitute -2 into the expression wherever there is an x, as follows:

$$2(-2)^3 - (-2)^2 + 3(-2) + 5$$
$$2(-8) - (4) + 3(-2) + 5$$
$$2(-8) - 4 + (-6) + 5$$

Recall that when taking negative numbers to odd powers, they retain their negative sign; however, when negative numbers are taken to even powers, they become positive. Thus the expression $2(-8) - 4 + (-6) + 5$ is equivalent to $-16 - 4 - 6 + 6 = 21$.

17. The correct answer is C. To find the next term in the geometric sequence, recall that when looking at a geometric sequence, the nth term can be written (for some base number b) as b^n. This sequence is decreasing in magnitude, thus the absolute value of b is less than 1. It also alternates signs, which indicates that b is negative. At this point, mathematical intuition might suggest trying $b = -\frac{1}{3}$, and then calculating for what value of n the terms in the given sequence correspond (see table).

Table 1

n	1	2	3	4	5
nth term	9	-3	1	$-\frac{1}{3}$	$\frac{1}{9}$
	$\left(-\frac{1}{3}\right)^{-2}$	$\left(-\frac{1}{3}\right)^{-1}$	$\left(-\frac{1}{3}\right)^{0}$	$\left(-\frac{1}{3}\right)^{1}$	$\left(-\frac{1}{3}\right)^{2}$

18. The correct answer is J. Since $\frac{1}{4}$ inch on the blueprint represents an actual length of 1 foot, 1 inch on the blue print represents an actual length of 4 feet. When the room is 3 inches by $4\frac{1}{4}$ inches on the blueprint, it is actually 4(3) feet by $4\left(4\frac{1}{4}\right)$ feet. Because the room is rectangular, the area is equal to length times width, so the area is $4(3) \times 4\left(4\frac{1}{4}\right) = 12 \times 17 = 204$ square feet.

19. The correct answer is A. Although the exact values of m and n are not known, it can be determined that the difference of m and n is always positive. This is true because subtracting a negative number is the same as adding the positive. Since $n < 0$, subtracting the negative n will always be a net addition to the value of m, which is a positive number to begin with. Therefore, answer choice A is correct.

20. The correct answer is G. To solve the equation $x + \frac{2}{3} = \frac{8}{21}$ for x, subtract $\frac{2}{3}$ from both sides.

Doing so will yield $x = \frac{8}{21} - \frac{2}{3}$. Fractions must have common denominators in order to be added or subtracted from one another. To obtain a common denominator in this problem, you can multiply $\frac{2}{3}$ by $\frac{7}{7}$ $\left(\text{multiplying by } \frac{7}{7} \text{ is like multiplying by 1, thus it does not change the value of the fraction}\right)$. $\frac{2}{3} \times \frac{7}{7} = \frac{14}{21}$. The equation is now $x = \frac{8}{21} - \frac{14}{21} = \frac{-6}{21}$, which reduces to $\frac{-2}{7}$.

21. The correct answer is C. The line $3x + 4y = -12$ is given in standard form. To find the slope, convert the equation into slope-intercept form ($y = mx + b$, where m is the slope and b is the y-intercept):

$$(3x + 4y) - 3x = -12 - 3x$$

$$\frac{(4y)}{4} = \frac{(-3x - 12)}{4}$$

$$y = \left(\frac{-3}{4}\right)x - 3$$

Since the equation is now in slope-intercept form, it is easy to see that the slope is $\frac{-3}{4}$.

22. The correct answer is K. A square with sides $(3x - 2)$ inches would have an area of $(3x - 2)(3x - 2)$ square inches. Use the "FOIL" method to multiply all terms in the equation:

First: $(3x)(3x) = 9x^2$

Outside: $(3x)(-2) = -6x$

Inside: $(-2)(3x) = -6x$

Last: $(-2)(-2) = 4$

Adding all of these terms yields the expression $9x^2 - 6x - 6x + 4$. After combining like terms, the expression is $9x^2 - 12x + 4$.

23. The correct answer is B. To solve this problem it is necessary to factor the polynomial $x^2 - 2x - 24$. In this case, since the final term is negative, you know that when this equation is factored, one sign will be positive and the other will be negative. Further, since the term is -24, the numbers in the problem will be factors of 24. The factors of 24 are 1 and 24, 2 and 12, 3 and 8, or 4 and 6. It is logical to look for the factors that have a difference of 2 because of the $-2x$ term in the original polynomial. In this case, 4 and 6 are the only factors of 24 that have a difference of 2. Because the term $-2x$ is negative, the logical expression is $(x + 4)(x - 6)$. Therefore, $x + 4$ is the correct answer choice.

24. The correct answer is H. As k gets larger and larger without bound, the expression $\frac{4}{(2 + k)}$ becomes 4 divided by an increasingly large number. For example, think about the trend between the following fractions: $\frac{4}{100}$, $\frac{4}{10,000}$, $\frac{4}{1,000,000}$, ... Looking at it this way, you can see that the expression for r gets closer and closer to zero.

25. The correct answer is D. To convert 25°C to degrees Fahrenheit using the formula $\text{F} = \left(\frac{9}{5}\right)\text{C} + 32$, simply substitute 25 for C in the formula:

$$\left(\frac{9}{5}\right)25 + 32 = 77°\text{F}$$

26. The correct answer is G. In a rectangle with length l which is 5 inches longer than the width w, the relation between the sides is $l = w + 5$. Since perimeter, which is the distance around, can be written as twice the length plus twice the width, $38 = 2l + 2w$. In order to solve for width w, use the equation $l = w + 5$ to substitute into the perimeter equation:

$$38 = 2(w + 5) + 2w$$

$$38 = 2w + 10 + 2w$$

$$38 - 10 = 4w + 10 - 10$$

$$\frac{28}{4} = \frac{4w}{4}$$

$$w = 7$$

27. The correct answer is D. For the equation $3x^2 - 2x - 21 = 0$, all solutions of x can be found by either factoring or by using the quadratic formula $\left[x = \frac{\left(-b \pm \sqrt{(b^2 - 4ac)}\right)}{2a}\right.$ for quadratic equations $\left.ax^2 + bx + c = 0\right]$. In this case, since the final term is negative, you know that when this equation is factored, one sign will be positive and the other will be negative. Knowing that the factors of 21 are 1 and 21 and 3 and 7, you can guess and check your answers with relative ease. In this case, the correct answer is $(3x + 7)(x - 3) = 0$. Solve for x: $x = \frac{-7}{3}$ or $x = 3$.

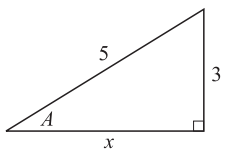
28. The correct answer is G. A good way to go about solving this problem is to think of the tests as each having a value of 100 points, making the total of all 3 tests worth 300 points. In order to average 87% overall, Sulema would then need to score 87% of the 300 points, or 300(0.87), or 261 points. By scoring 88% and 79% on her previous tests, she has already obtained 88 + 79, or 167 points. Thus Sulema must score 261 − 167, or 94 points on the third test, which is equivalent to 94%.

29. The correct answer is A. If $a^b = m$ and $c^{2b} = n$, then $mn = (a^b)(c^{2b})$. Because a and c are not necessarily the same number, you cannot simply add the exponents. The only legitimate answer choice that further manipulates the expression $(a^b)(c^{2b})$ factors out the b exponent and places it outside a set of parentheses. Thus $(a^b)(c^{2b}) = (a^b c^{2b}) = (ac^2)^b$.

30. The correct answer is J. A circle with diameter 12 inches has a radius of 6 inches. The area of a circle is $A = \pi r^2$, where r is the radius. Thus, the area of this circle is $A = \pi(6^2) = 36\pi$ square inches.

31. The correct answer is E. The percent of students who pass both exams is (85%)(70%), or 0.85×0.70, which is 0.595. In a sample of 100, 100×0.595, or 59.5 (roughly 60) students would be expected to pass both exams.

32. The correct answer is G. Given that $\sin A = \frac{3}{5}$, you might want to draw a picture like the one below to illustrate the situation.



In this case, the ratio of the side opposite angle A to the hypotenuse is $\frac{3}{5}$. To find $\tan A$, the length of the other leg is needed. This can be obtained using the Pythagorean Theorem, or by knowing that this is a special 3–4–5 right triangle. According to the Pythagorean Theorem (in a right triangle with sides a, b, and c, where c is the hypotenuse, $c^2 = a^2 + b^2$), $5^2 = 3^2 + x^2$, where x is the length of the unknown leg. Solve for x:

$$x^2 = 5^2 - 3^2$$

$$x^2 = 25 - 9$$

$$x^2 = 16$$

$$x = 4$$

Once the length of the third leg is obtained, $\tan A = \frac{\text{opposite}}{\text{adjacent}} = \frac{3}{4}$.

33. The correct answer is D. To solve $\frac{[(x-3)(x-6)]}{[(3-x)(x-6)]}$ for any x other than 3 or 6, first try to cancel any possible quantities. In this case, $(x - 6)$ appears in both the numerator and the denominator, so it can be canceled out of the equation. The new equation is then $\frac{(x-3)}{(3-x)}$. You also must realize that $(3 - x)$ can be written as $-1(x - 3)$. Thus the remaining expression is:

$$\frac{(x-3)}{(-1(x-3))}$$

$$-1 \times \frac{(x-3)}{(x-3)}$$

$$-1 \times 1 = -1$$

34. The correct answer is G. The first step in solving this problem is to factor $\sqrt{27}$ and $\sqrt{48}$:

$$\sqrt{27} = \sqrt{9} \times \sqrt{3} = 3\sqrt{3}$$

$$\sqrt{48} = \sqrt{16} \times \sqrt{3} = 4\sqrt{3}$$

Now, you can add the simplified numbers; thus $\sqrt{27} + \sqrt{48} = 3\sqrt{3} + 4\sqrt{3}$, or $7\sqrt{3}$.

35. The correct answer is C. When AB is 5 inches long, BC is 8 inches long, and AC is 3 inches long, $\triangle ABC$ has a perimeter of $5 + 8 + 3 = 16$. To find the perimeter of $\triangle XYZ$ when its longest side is 20, you can set up proportions to find the lengths of the sides, since the sides of similar triangles are in proportion to each other. In this case, you can take the proportion of the longest sides, 20:8, and apply it to the perimeter p:

$$\frac{20}{8} = \frac{p}{16}$$

$$p = \frac{20 \times 16}{8} = 40$$

36. The correct answer is H. When square *ABCD* is divided as shown, each of the small triangles has legs of length 2 and 4. They also happen to be right triangles, because each shares one angle with the corner of the large square, which are right angles. To solve for the area of the shaded region, you can use the Pythagorean Theorem to calculate the length of the side of the shaded square. Alternatively, you can find the area of one triangle, quadruple that value, and subtract it from the area of square *ABCD*, which is 6^2, or 36 square inches.

The area of a triangle is equal to $\frac{1}{2}$ (base)(height). In a right triangle, either leg will work as the value for the base or the height, so the area of one triangle is $A = \frac{1}{2}(2)(4) = 4$. The area of the 4 triangles together, then, is 4×4, or 16. Thus the area of the shaded region is $36 - 16$, or 20.

37. The correct answer is B. Since *AC* is the diameter of the circle, and *D* is the midpoint of *AC*, *D* is the center of the circle. Thus the distance from *D* to any point on the circle is equal to the radius, *r*. Since *B* is a point on the circle, the distance *BD* is equal to *r*. Since *A* is a point on the circle, the distance *AD* is also equal to *r*. Further, since *AB* is congruent to *BC*, the segment *AB* is perpendicular to the segment *AD*, thus the measure of angle *ADB* is 90°. In the triangle *ABD*, two sides are equal to *r*, thus the triangle is isosceles with one of the angles equal to 90°. The other two angles are then equal in measure and add up to 90°. Therefore, angle *ABD* has a measure of 45°.

38. The correct answer is J. To solve this problem, recall that the midpoint formula for points (x_1, y_1) and (x_2, y_2) is $\left(\frac{(x_1 + x_2)}{2}, \frac{(y_1 + y_2)}{2}\right)$. Therefore, for points (−1,3) and (2,5), the midpoint is $\left(\frac{(-1+2)}{2}, \frac{(3+5)}{2}\right) = \left(\frac{1}{2}, \frac{8}{2}\right)$, or $\left(\frac{1}{2}, 4\right)$.

39. The correct answer is C. Maria ran 5 miles in 37 minutes 29 seconds, which is about 37.5 minutes. To solve for miles per hour, first calculate how long it took Maria to travel one mile. $\frac{37.5 \text{ minutes}}{5 \text{ miles}} = \frac{7.5 \text{ minutes}}{\text{mile}}$, or 1 mile in 7.5 minutes. To convert this into miles per hour, simply multiply by the number of minutes there are in one hour: $\left(\frac{60 \text{ minutes}}{1 \text{ hour}}\right) \times \left(\frac{1 \text{ mile}}{7.5 \text{ minutes}}\right) = 8$ miles per hour.

40. The correct answer is J. To find $g(f(x))$ using the values of the functions $f(x)$ and $g(x)$ as given in the tables, think of $g(f(x))$ as applying the function g to $f(x)$. So, first find $f(x)$ for $x = -1$; $f(-1) = 2$. Then, find $g(x)$ for $x = 2 : g(2) = 3$.

41. The correct answer is D. To find an equivalent expression for $x^{\frac{1}{2}}y^{\frac{3}{4}}z^{\frac{5}{8}}$, it is important to be familiar with exponent rules. For example, $x^{\frac{2}{3}} = \left(x^2\right)^{\frac{1}{3}} = \sqrt[3]{x^2}$. The first step in this problem would be to "pull out" a common multiple from the exponents. To better see this, it might be helpful to convert all of the exponents into a common denominator: $x^{\frac{1}{2}}y^{\frac{3}{4}}z^{\frac{5}{8}} = x^{\frac{4}{8}}y^{\frac{6}{8}}z^{\frac{5}{8}}$. In this case, you can see that $\frac{1}{8}$ is the common multiple: $x^{\frac{4}{8}}y^{\frac{6}{8}}z^{\frac{5}{8}} = \left(x^4y^6z^5\right)^{\frac{1}{8}}$. Further, $\left(x^4y^6z^5\right)^{\frac{1}{8}}$ can be written as $\sqrt[8]{x^4y^6z^5}$.

42. The correct answer is F. To solve $A = \frac{1}{2}h(b_1 + b_2)$ for b_1, divide both sides by h and multiply both sides by 2. The remaining equation is $2\frac{A}{h} = b_1 + b_2$. Then simply subtract b_2 from both sides of the equation to get $b_1 = 2\frac{A}{h} - b_2$.

43. The correct answer is B. To find the rate of gasoline use, which translates into miles per gallon, divide the number of miles traveled by the number of gallons of gasoline used. In this case, you can see that the graph goes through the point (100,5), (200,10) and so on. Based on this information, the rate would be $\frac{100 \text{ miles}}{5 \text{ gallons}}$, or 20 miles per gallon.

44. The correct answer is K. In a graph, the values at which $y = 0$ will be where the graph of a function crosses the *x*-axis. As seen in the figure, the graph of $y = ax^2 + bx + c$ crosses the *x*-axis twice; thus there will be two real solutions for the equation $0 = ax^2 + bx + c$. Imaginary solutions would apply if the graph did not cross the *x*-axis at any point.

45. The correct answer is B. Recall that $|x|$ means "absolute value of *x*," which is the magnitude of a number, regardless of its sign. For instance $|-2| = |2| = 2$. In the case of this problem, the equation $|x| = x + 12$ can be broken into 2 related sub-equations: $x = x + 12$ and $-x = x + 12$. When

$x = x + 12$, the equation does not make sense because if you subtract x from both sides, $0 = 12$ remains, which obviously is false. With the other sub-equation, $-x = x + 12$, you can subtract an x from both sides to get $-2x = 12$. Dividing by -2, you arrive at $x = -6$.

46. The correct answer is H. To solve this problem, convert $\frac{1}{3}$ and $\frac{1}{2}$ to fractions with a common denominator, such that $\frac{1}{3} = \frac{2}{6}$ and $\frac{1}{2} = \frac{3}{6}$. The halfway point will be somewhere between $\frac{2}{6}$ and $\frac{3}{6}$. From there, you can increase the magnitude of the denominator so that a halfway point between the numerators is more evident. $\frac{1}{3} = \frac{2}{6} = \frac{4}{12}$ and $\frac{1}{2} = \frac{3}{6} = \frac{6}{12}$. Thus the halfway point is between $\frac{4}{12}$ and $\frac{6}{12}$, which is $\frac{5}{12}$.

47. The correct answer is E. Because Laura recorded the \$20 as income, the balance of the budget would appear to be \$20 higher than it should be. However, because the \$20 was an expense, once the \$20 was deducted from the income statement and then deducted again from the expense report, the balance would appear to be \$40 more than it should.

48. The correct answer is K. Think of each shift as a spot to be filled: __×__×__×__×__. In each spot there are only a certain number of available workers. For example, the first spot can have any of the 5 volunteers working, leaving 4 possible workers for the next shift, and so on. Therefore, the number of possible arrangements of volunteers is $5 \times 4 \times 3 \times 2 \times 1 = 5!$, or 120.

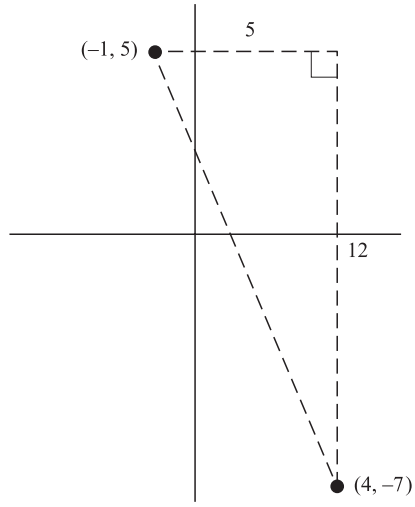
49. The correct answer is C. To find the distance between points (4,−7) and (−1,5) in the standard (x,y) coordinate plane, use the distance formula $d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$ as follows:

$$\sqrt{(-1-4)^2 + (5+7)^2}$$

$$\sqrt{(-5)^2 + (12)^2} = \sqrt{169}$$

$$\sqrt{169} = 13$$

Another way to look at this problem would be to draw a picture (as below). Then you can use the Pythagorean Theorem to find the hypotenuse of the triangle that is formed from those points.



50. The correct answer is G. The diameter of the cylinder is 3 meters. Therefore, the radius is 1.5 meters. Using the formula for the volume of a right circular cylinder, $V = \pi r^2 h$ with h equal to the length 10 meters, $V = \pi(1.5)^2(10)$, or approximately 70.65 cubic meters. Since water weighs about 2,205 pounds per cubic meter, the weight of the water cargo is $2{,}205 \times 70.65$, or about 156,000 pounds.

51. The correct answer is C. Since the two lines are parallel, the two triangles are similar and share the same constant ratio. The smaller triangle has one side with a length of 4 cm, and the larger triangle has one side with a length of $4 + 12$, or 16 cm. The legs of the triangles have a ratio of 4:16, which can be reduced to 1:4. The ratio of 1:4 also extends to the area of the triangles. You are given that the smaller triangle has an area of 8cm^2. The larger triangle has an area that is 4 times larger, so the area of the larger triangle is $4 \times 8 = 32\ \text{cm}^2$.

52. The correct answer is G. To solve for the total measure of all interior angles of a regular polygon, use the formula $(n-2)(180°)$ where n is the number of vertices of the polygon. A regular hexagon has 6 vertices, so the total measure of all interior angles is $(6-2)(180) = (4)(180) = 720°$. Since there are six interior angles, each one is $\frac{720°}{6} = 120°$.

53. The correct answer is B. The best way to solve this problem is to set up a table indicating the time

period in years and the number of both CDs and video games purchased during the years given. The rate of purchase is the same, so, based on information in the problem, you can fill in the table below.

Time period	CDs	Video games
1993	7	6
1994	9	9
1995	11	12
1996	13	15
1997	15	18

Teenagers bought the same average number of CDs and video games in 1994.

54. The correct answer is K. The first step in answering this question is to set $x^2 - 45b^2 = 4xb$ equal to zero, by isolating all of the terms on the left side:

$$x^2 - 4xb - 45b^2 = 0$$

The next step is to factor the equation and set each group equal to 0 in order to solve for x:

$$(x + 5b)(x - 9b) = 0$$

$$x + 5b = 0, \text{ so } x = -5b$$

$$x - 9b = 0, \text{ so } x = 9b$$

The two solutions for x are $-5b$ or $9b$.

55. The correct answer is D. This question tests your ability to recognize and apply the distributive property of multiplication. According to the distributive property, for any numbers a, b, and c, $a(b - c) = ab - ac$.

$a(b - c) = ab - ac$, which is NOT equivalent to $ca - ba$, so Roman Numeral I is incorrect; eliminate answer choices B, C, and E.

ab - ac, so Roman Numeral II is correct.

$(b - c)a$, so Roman Numeral III is also correct.

Since II and III are equivalent to $a(b - c)$, answer choice D is correct.

56. The correct answer is F. By definition, the tangent of any angle is the $\frac{\sin}{\cos}$. Therefore, $\frac{(\cos x)}{(\tan x)(\sin x)}$ is equal to $\frac{\cos x}{\frac{(\sin x)}{(\cos x)}(\sin x)}$. Multiply both the numerator and denominator by $\cos x$ to get $\frac{\cos^2 x}{\sin^2 x}$.

57. The correct answer is E. By definition, if a and b are inversely proportional, then $a_1b_1 = a_2b_2$. Therefore, $(36)(12) = (48)a$. Solve for a as follows:

$$(36)(12) = (48)a$$

$$432 = 48a$$

$$9 = a$$

58. The correct answer is F. The best way to solve this problem is to plug in the answer choices for c, and factor the equation, starting with answer choice F:

$$5x^2 + 16x + 3 = 0$$

$$(5x + 1)(x + 3) = 0$$

$$5x + 1 = 0, x = \frac{-1}{5}$$

$$x + 3 = 0; x = -3$$

Answer choice F gives you 2 distinct solutions for x, so you do not need to try the remaining answer choices. Using the quadratic formula to solve the other choices will not give you real solutions.

59. The correct answer is A. To solve this problem, you must realize that the volume of a cube is equal to (length)(width)(height) or simply $(\text{side})^3$, since all sides of a cube are equivalent in length. To find the length of one side, find the cube root of 64, which is 4. $(4^3 = 64)$ Because all sides of a cube are equal, the shortest distance from the center of the cube to the base of the cube will equal the midpoint of the length of the cube, which is $\frac{4}{2}$, or 2.

60. The correct answer is K. The line $5x + 8y = 17$ is given in standard form. You should find the slope of this line by converting the equation into

slope-intercept form ($y = mx + b$, where m is the slope and b is the y-intercept):

$$5x + 8y = 17$$
$$8y = -5x + 17$$
$$y = \frac{-5x}{8} + \frac{17}{8}$$

Since the equation is now in slope-intercept form, it is easy to see that the slope of the line is $\frac{-5}{8}$. The slope of a line perpendicular to the line with a slope of $\frac{-5}{8}$ is $\frac{8}{5}$, answer choice K (remember that perpendicular lines have negative reciprocal slopes).

Reading Test Explanations

PASSAGE I

1. **The best answer is A.** In the second paragraph, the narrator states, "I didn't want to start a trend of hair-raising stunts," but his father continues with his plan for a rappelling platform anyway.

2. **The best answer is J.** The passage focuses on the fun that the narrator, his father, and his friends had while building a rappelling platform, even though there was some hesitation on the part of the narrator. Only the first paragraph details the narrator's father's popularity, and nothing in the passage suggests that the narrator is "struggling" with it.

3. **The best answer is C.** The first and second paragraph demonstrate that the narrator is more timid than his father when it comes to exhilarating stunts. The narrator is not sure whether he wants to bring his friends to his father for ideas.

4. **The best answer is J.** The first sentence of the passage establishes a timeframe that reveals the narrator is no longer a child: "As I was growing up, . . ." Therefore eliminate answer choices G and H. The tone of the passage is positive so answer choice J is best.

5. **The best answer is A.** The last sentence of the first paragraph describes the narrator's father's "extreme career and energetic lifestyle." Answer choice D might seem correct, although the passage notes that the narrator's father was once a Hollywood stuntman, indicating that he would be trained in safety procedures, some of which arise during his "impromptu lesson on the critical safety rules."

6. **The best answer is H.** In the paragraph, the author describes the pleasure of following his father around the lumberyard, as well as the anticipation of what all the "shiny, new parts" would serve to build, which best supports answer choice H.

7. **The best answer is B.** After this sentence, the rest of the passage describes how the narrator's friends convinced him to consult his father for advice on a bigger, bolder stunt. This information suggests that this uncommon adventure, at least, was the result of the narrator's father's unusual occupation.

8. **The best answer is J.** According to the passage, "My father is recently retired from his career as a Hollywood stunt performer. When I was a child, he would do more death-defying tricks in a week than I'm sure I will ever do in my lifetime. My father's extreme career and energetic lifestyle made him the coolest dad in town, and I had to live up to him," which best supports answer choice J.

9. **The best answer is D.** The word "rigging" is a well-known sailing term, referring to the ropes and pulleys used with the sails. By extension, any support system of ropes or similar lines—such as the one required for rappelling—could be called "rigging." The other answer choices are mentioned in the passage, but do not have the same meaning as "rigging."

10. **The best answer is G.** The passage states that "At first, I resisted, since I didn't want to start a trend of hair-raising stunts on my neighborhood street. Who knows what the old lady across the street would think?" Answer choice G is the best choice because while the passage focuses on a fun experience that the narrator had with his friends, it also describes his apprehension to do anything that could be unreasonably dangerous or disruptive to the neighborhood.

PASSAGE II

11. **The best answer is B.** The passage states that Robert Catesby, Guy Fawkes, and several other Roman Catholics "were thought to be denouncers of the king's own Church of England," and they "decided that unless something was done from within, nothing would likely change." These statements suggest that some English Roman Catholics did not agree with the king and rebelled against him, seeking their own religious freedom. The other answer choices are not supported by the passage.

12. **The best answer is H.** The passage asserts that the plan was carefully conceived and well executed, except just before the planned detonation, when a conflicted conscience led one conspirator to reveal the plan to a government official, which best supports answer choice H.

13. **The best answer is A.** The last paragraph states that "Guy Fawkes Day serves as a chilling reminder" for British people. Answer choice D seems plausible, but the passage does not indicate that modern British citizens find fault with (i.e., consider depraved) Fawkes' reaction to the king's policy governing religion.

14. **The best answer is J.** The phrase "tongue-in-cheek" implies humor at the expense of members

of parliament. Skepticism of the true motives of legislators is evident in the passage.

15. **The best answer is C.** The final two sentences of the passage state that "an individual will press back . . . in defense of his beliefs." This best supports answer choice C.

16. **The best answer is H.** The execution was meant to teach a lesson to the public. The king wanted to make a powerful statement against treason, so he chose an "unusual" punishment to shock the masses. The word "exotic" can refer to something that is "unusual" or "unexpected," so it is the best choice here.

17. **The best answer is A.** As Fawkes and his Catholic cohorts saw it, the king was unreasonably treading on their right to practice their own religion. In the second paragraph, the author asserts that the Church of England was "the king's own," insinuating that it held political sway. Hence, answer choice A is correct because it represents an action of government taken to preserve its power. The other choices reflect government actions taken for the common good.

18. **The best answer is G.** The passage implies that although the threat of punishment may deter most people from action, "no brutal threat can stop the most committed believer," which best supports answer choice G.

19. **The best answer is A.** The passage states that the conspirators were lucky to find a cellar space beneath Westminster Palace. A physical sense of "stumbled upon," as in answer choices B and C does not make sense in context. Answer choice D is incorrect because the decision to seek something seems unrelated to luck.

20. **The best answer is H.** According to the passage, the conspirators were preparing an explosion underneath the floor of Parliament in order to maximize destruction. The "mineshaft" refers to the actual space to be filled with explosives.

PASSAGE III

21. **The best answer is A.** The first two paragraphs describe the qualities of Southern Gothic literature. The rest of the passage relates Tennessee Williams' tumultuous life story, explaining in the process how the author's family members inspired several of the his characters, which best supports answer choice A.

22. **The best answer is G.** The passage maintains an even, analytical tone, meaning the author does not attempt to persuade the reader to feel a certain way about the facts of Williams' life; the author is interested in William's life, but remains detached, or apart, from it. The other answer choices suggest emotions that are not indicated by the tone of the passage.

23. **The best answer is B.** The year of publication of Williams' plays mentioned in the passage are (in order of appearance) 1948, 1955, 1944, and 1958. The passage states that the play from 1948, *A Streetcar Named Desire*, won a Pulitzer Prize, perhaps the most prestigious literary award in the United States.

24. **The best answer is H.** The prefrontal lobotomy is described as a "now-defunct" ("outdated") procedure that was "ruinous" ("destructive") to Rose, rendering her a vegetable. The other choices hint at some benefit to the procedure, but the passage asserts that no good to Rose nor to the Williams family came from the treatment.

25. **The best answer is C.** The third paragraph describes how Williams used family members as models for his characters. The final paragraph details how emotionally damaged he was for the rest of his life. This best supports answer choice C. Answer choice D may be distracting, but the fact that Williams is "among the most celebrated" of Southern Gothic playwrights proves that his work did indeed appeal to his audience.

26. **The best answer is J.** The phrase "blatant accusations" refers to transparent rhetoric that might come from a character taking a stand on the "moral issues" of the past. The richness of Southern Gothic literature is due in part to the absence of overt commentary on such subjects. The reader is left to infer the morality in the prose from the actions of the characters. The word "overtly" refers to something that is open and obvious.

27. **The best answer is C.** The passage states that Williams' "mother . . . pushed him toward creative arts during his period of illness." He "heartily accepted" her gift of a typewriter. These statements best support answer choice C.

28. **The best answer is F.** According to the passage, Southern Gothic literature "reflects life in the American South. It maintains some of the characteristics of Gothic writing, such as use of the supernatural or the ironic; however, Southern Gothic does not focus on creating tension and suspense as do other Gothic genres." In addition, the

passage states that "that the Southern Gothic writer is able to present and explore moral issues of the American South, such as slavery and bigotry . . ." These statements best support answer choice F.

29. **The best answer is C.** A person "depressed and fighting alcoholism" can be said to be "embroiled" in personal problems. As for Williams' characters, which are characteristic of Southern Gothic style, they are brooding and often uninviting, and it is through them that "moral quandaries" (ethical dilemmas) are analyzed.

30. **The correct answer is F.** The word "postbellum" means, in general, "post-war," but it is usually restricted to mean "post-Civil-War." With the end of the Civil War came the end of slavery, but antebellum, or pre-war attitudes of "racism and inequality" persisted, which are what Williams' characters struggle to reconcile. This best supports answer choice F.

PASSAGE IV

31. **The best answer is B.** According to the passage, jack pine "depends on regular fires to expose the seeds within its tough cones." It is likely that any management of the forests would include periodic, supervised burning in order to encourage new growth.

32. **The best answer is F.** The passage states that, "The early 1900s, though, saw natural fires suppressed by new forest management policies and, consequently, jack pine forests quickly diminished." These fire prevention policies stopped all fires, most likely because of the destructive threat they pose to land, buildings, and infrastructure. An unforeseen consequence ("externality") was the depletion of fire-dependent jack pines, which constitute the exclusive habitat of the Kirtland's warbler.

33. **The best answer is D.** The passage states that the warblers are "extremely fastidious [picky] about their habitat." Subsequent paragraphs describe historical strain on Jack pine forests which host the birds. Answer choices B and C are never said to be detrimental to the success of the bird. Answer choice A, while detrimental, is coincidental to the birds' renewed proliferation among the jack pines. The principal reason that the bird is endangered is the widespread decline of suitable habitat.

34. **The best answer is J.** The passage states that the Kirtland's warbler "migrates to the Bahamas to winter." This suggests a temporary stay, much like how people in the United States may winter in Florida, but still call their northern residence home. The warbler engages in its most complex behavior while in Michigan (i.e., building nests and mating), which best supports answer choice J.

35. **The best answer is C.** According to the passage, forest policy at the turn of the century dictated that all fires should be prevented; as a result, Jack pines, which require fire to reproduce, suffered, which best supports answer choice C.

36. **The best answer is F.** The word "brood" describes the group of eggs a bird raises at one time. Thus a "brood parasite" is an organism that preys on the nestling of another bird.

37. **The best answer is B.** The fact that the passage claims that Kirtland's warblers are "fastidious about their habitat" indicates that their habitat requirements are uncommonly rigid as compared to other birds, which best supports answer choice B.

38. **The best answer is F.** The passage states "the tree depends on regular fires to expose the seed within its tough cones." This indicates the cones are ready prior to the fire, eliminating answer choice G. Answer choices H and J are not supported by the passage.

39. **The best answer is C.** The Jack pine problem is cited as the major contributing factor to Kirtland's warbler decline. These trees are the bird's habitat. Answer choice D might appear to be correct, but the logging of jack pines is not cited as a reason for the widespread depletion of the tree; fire prevention is.

40. **The best answer is G.** The word "percolates" means "drains" in the context of soil, indicating it is "permeable." This is important to prevent flooding of nests. The passage continues by indicating that Grayling sand supports the plants that the Kirtland's warbler requires.

Science and Reasoning Test Explanations

PASSAGE I

1. **The best answer is C.** According to Figure 2, Amino Acid A (the solid line) has the highest rate (on the *y*-axis) at a water volume between 10 mL and 15 mL. This volume is closest to answer choice C, 12 mL.

2. **The best answer is H.** Amino Acid A and Amino Acid B have the same reaction rate when the two lines cross on the graph. This happens at a water volume of approximately 50 mL.

3. **The best answer is A.** Figure 3 shows that the reaction rate of Amino Acid B is not dependent on the residue concentration. Figure 4 shows that the reaction rate of Amino Acid B is dependent on the amino acid concentration. Therefore, the figures do not support the researcher's claims that the reaction of Amino Acid B is dependent on both concentrations.

4. **The best answer is G.** According to Figure 4, Amino Acid A always has a lower rate of reaction than Amino Acid B. This does not support the researcher's claims that the reaction rate for Amino Acid A at any given concentration will always be greater than Amino Acid B for the same concentration. Answer choices F and H can be eliminated because the effect of residue concentrations on the rate of reaction is not shown in Figure 4.

5. **The best answer is A.** According to the figure, the water volume of peak reaction rates for amino acids in the muscles (glycine) is higher (35 mL) than that of amino acids in DNA (alanine–20 mL).

PASSAGE II

6. **The best answer is F.** Scientist 1 discusses nature as the major factor contributing to the low lake levels. Scientist 2 discusses human activities as the major contributing factor. This is the major difference between the scientists' viewpoints. Answer choices G and H can be eliminated because the scientists do not discuss the effects of the water or lake levels.

7. **The best answer is C.** The two scientists are discussing the causes of low lake levels in the Great Lakes. This implies that both scientists agree that recent trends show that the lake levels are decreasing. Answer choice B can be eliminated because only Scientist 2 supports this viewpoint.

8. **The best answer is J.** Scientist 1 says, "the influence of human regulation on lake levels is inconsequential. Because water is added through snow and rain and taken away through evaporation, nature has most of the control over lake levels." The scientist believes that the lake levels are mainly controlled by nature, and human manipulation does not interfere. Answer choices F and H can be eliminated because only Scientist 2 discusses this information.

9. **The best answer is B.** Scientist 2 talks about a number of ways that human activity causes the Great Lakes' water levels to decrease. Activities such as the Lake Michigan diversion at Chicago, the dredging and modifying of the St. Clair and Detroit rivers, and channeling and shore lining modifications in connecting the channels of the Great Lakes have all contributed to the low lake levels. This information best supports answer choice B.

10. **The best answer is G.** If studies have shown precipitation and evaporation levels to be stable for the last 50 years, Scientist 1's viewpoint would be weakened. This is because Scientist 1 says a recent increase in evaporation has contributed to lower lake levels: "The recent decline of water levels in the Great Lakes, now at lows not seen since the mid-1960s, is due to a number of causes. High degrees of evaporation from warmer than usual temperatures in recent years . . ." Because this statement would be disproved, the scientist's viewpoint would be weakened.

11. **The best answer is A.** The best way to answer this question is by the process of elimination. Answer choice B is incorrect because the passage says, "the Lake Michigan diversion at Chicago moves water out of Lake Michigan and into the Mississippi River for domestic, navigation, hydroelectric, and sanitation purposes." Answer choice C is incorrect because the passage states "the St. Clair and Detroit rivers have been dredged and modified. This has caused some drop in the levels of Lake Michigan and Lake Huron." Answer choice D is incorrect because the passage says, "the infilling of shoreline areas can reduce the flow carrying capacity of the river." Answer choice A is correct because it is the only answer choice that is not discussed by Scientist 2.

12. **The best answer is G.** If the water deposited in aquifers surrounding the Great Lakes was used within the Great Lakes' watershed, the water would be cycling through the Great Lakes, and no water would be taken outside of the cycle. Thus, this would have little or no effect on lake levels, answer choice G.

PASSAGE III

13. **The best answer is A.** If heat were to flow from the water in the capsule to the surrounding oven, the heat flow would be negative because the water is losing energy. The only trial in which this occurred is Trial I, answer choice A (Trial 1: −94 J/s).

14. **The best answer is H.** If the best insulators are those that are the poorest heat conductors, material with the lowest heat flow are the best insulators. Of the four answer choices, fiberglass has the lowest heat flow, so it is the best insulator.

15. **The best answer is D.** According to Table 1, heat flowed at a rate of 59 J/s when 1.0 inch of cellulose was used as an insulator. This is the greatest flow rate among the combinations shown in the answer choices.

16. **The best answer is H.** The thermal conductivity for fiberglass at 1.0 in. and 1.5 in. is 38 J/s and 32 J/s, respectively. The thermal conductivity for cellulose at 1.0 in. and 1.5 in. is 59 and 45 J/s, respectively. The results show that cellulose has a higher conductivity than fiberglass and proves answer choice H is NOT true. Therefore, answer choice H is the correct answer.

17. **The best answer is D.** Trials 3 and 5 both had the same insulation material, initial temperature, and oven temperature; however, they had different heat flows. The only condition that was varied between the two trials was the thickness of the insulator. This suggests that heat flow depends on the thickness of the insulating material, answer choice D.

PASSAGE IV

18. **The best answer is G.** According to Tables 1 and 2, as the weight of the object increases (going down the column) for each brand, the average force required to move the object also increases. In other words, there is a direct relationship.

19. **The best answer is C.** Table 1 shows that for brand B cookware, when the mass of the weighted object was 150 g and 250 g, the average force required to move the object was 0.090 N and 0.147 N, respectively. Because 200 g is in between 150 g and 250 g, you would expect the force needed to move an object that has a mass of 200 g to be between 0.090 and 0.147 N. Answer choice C (0.12 N) is the only answer choice within this range.

20. **The best answer is H.** The results from Table 1 show that brand C cookware has the smallest coefficient of static friction. The results from Table 2 show that brand Y cooking spray has the smallest coefficient of static friction. This information suggests that these two brands, if combined, would result in the surface with the least coefficient of static friction.

21. **The best answer is B.** Table 2 shows the results of Experiment 2. In the table, brands X and Y were both tested with 50 g, 150 g, and 250 g objects. Brand Z, however, was only tested with 150 g and 250 g objects. Answer choice B is correct because only brand Z cooking spray was tested with only 2 different weights in Experiment 2.

22. **The best answer is J.** The Experiment states: "The students planned to calculate the coefficient of static friction by determining the force required to disturb an object from rest." Answer choice H can be eliminated because the coefficient of static friction could be determined in both experiments, but cooking spray was only used in Experiment 2.

23. **The best answer is A.** According to Table 1, brand B cookware had an average force of 0.090 N for the 150 g weight and 0.147 N for the 250 g weight. These results are the closest to the results the students obtained when their instructor gave them the new piece of nonstick cookware.

PASSAGE V

24. **The best answer is F.** In Experiment 1, the control group (the group that the results are compared to) included the rosebushes that were grown in soil with no pesticide added.

25. **The best answer is D.** The best way to answer this question is by the process of elimination. Answer choices A and B are incorrect because they contradict information in the passage. Answer choice C is incorrect because this explanation is irrelevant to the topic of the passage. Answer choice D is the correct answer because if rosebushes generally die when pesticides are applied to them directly, it would make sense to apply to the pesticides to the soil.

26. The best answer is G. If there is a direct correlation between plant weight and the number of petals on the flowers, the lower the average weight of the rosebush, the lower the number of petals. A rosebush grown in Soil Type 2 would have the lowest number of petals with Pesticide A at 35 ppm (23.6 oz).

27. The best answer is D. If Pesticide C was used to treat a rosebush, no prediction could be made on the basis of the previous results because no information about Pesticide C was given. Answer choice C can be eliminated because the fragrance of the flowers was not tested in this experiment.

28. The best answer is G. The results of Experiment 2 are shown in Table 2. According to the table, the average weight of the rosebush was lowest in every row for Soil Type 1 and Pesticide A.

PASSAGE VI

29. The best answer is D. According to Table 1, the mass of the precipitate in Test Tube 1 was 4.3 mg. The difference between this test tube and the other test tubes is that it had a shorter incubation time (0 min) than other test tubes at the same temperature water bath.

30. The best answer is F. In Experiment 1, "7 milliliters of a *peptide* (a neurotransmitter) solution was added." In Experiment 2 "8 mL of peptide solution was added" to the test tubes. One difference between the two experiments is the amount of peptide solution used. A larger volume of the peptide solution was used in Experiment 2, answer choice F.

31. The best answer is C. According to Table 2, as the pH level increases (going down the table), the mass of the precipitate first increases from 2.5 mg to 6.2 mg and then starts to decrease. This information best supports answer choice C. Answer choice D can be eliminated because the masses of the precipitates in Table 2 do not remain constant.

32. The best answer is F. According to Table 1, the masses of the precipitate for Test Tube 1 (incubation period 0 min) and Test Tube 2 (incubation period 5 min) were 4.3 mg and 3.9 mg, respectively. If Tube 5 is incubated for 5 min and has a mass precipitate of 3.6 mg, the mass is just below that of Tube 2 for the same incubation period. It could be predicted that if Tube 5 were not incubated, the mass of its precipitate would equal 4.1 mg, just below that of Tube 1's mass.

33. The best answer is A. According to Table 1, 3.9 mg of precipitate was collected from Test tube 2 at a temperature of 25°C after a 5-minute incubation period. This is the greatest amount of precipitate yielded among the combinations shown in the answer choices.

34. The best answer is H. To answer this question, you must look at the results in Tables 1 and 2. According to Table 1, the mass of the precipitate is the least for test tubes with an incubation period of 15 min for 30°C (1.4 mg). According to Table 2, the mass of the precipitate is least for the test tube with a pH of 2.0. Therefore, if combined, these two conditions would likely produce the lease amount of precipitate.

PASSAGE VII

35. The best answer is D. According to Table 1, *Salmonella* bacteria had the greatest percent growth at a pH level of 5. This information suggests a pH level near 5 might best contribute to the growth of the bacteria.

36. The best answer is H. To answer this question you must look at the results shown in all of the tables and choose the conditions that create the greatest percent growth of bacteria. In Table 1, a pH level of 5 created the highest percent growth (81%) of the bacteria. In Table 2, the organic compound in Dish 1 created the greatest percent growth (37%) of the bacteria. In Table 3, a temperature of 40°C created the greatest percent growth (83%) bacteria. By combining these three conditions you would expect to get the greatest percent growth.

37. The best answer is A. Table 1 shows that the *Salmonella* bacteria reproduced most efficiently in a pH of 5 (acidic). A conclusion stating that *Salmonella* bacteria reproduce most efficiently in an acidic environment would reaffirm the results of Experiment 1.

38. The best answer is G. To maximize the length of time that bacteria will reproduce, the bacteria should be provided an unlimited supply of nutrients. In order to make sure the nutrients are not depleted, the bacteria groups must be regularly resupplied with unlimited nutrients, answer choice G.

39. The best answer is B. In a scientific experiment, the independent variable is the factor that is changed by the experimenter during the experiment. In Experiment 3, the temperature was changed; each dish had a different temperature. The factor being studied or observed, in this case the percent of the growth, is the dependent variable. Whether it changes depends on the manipulation of one or more independent variables.

40. The best answer is J. Answer choice J is the best answer because it is the only answer choice that tests the bacteria at intervals. Because bacteria reproduce at a rate that varies from one stage to the next, by testing at 15-min intervals you could study the different stages of growth most effectively.

Writing Test Explanation

Because grading the essay is subjective, we've chosen not to include any "graded" essays here. Your best bet is to have someone you trust, such as your personal tutor, read your essays and give you an honest critique. If you plan on grading your own essays, review the grading criteria and be as honest as possible regarding the structure, development, organization, technique, and appropriateness of your writing. Focus on your weak areas and continue to practice in order to improve your writing skills.

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